WASP-43b

R-0599

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_230219` · created 2026-07-25T03:09:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459994.8373 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.058
Transit depth (Rp/Rs)^2	3.91 +/- 2.31 [%]
Orbital Inclination (inc)	79.99 +/- 2.94
Airmass coefficient 1 (a1)	6.03 +/- 2.92
Airmass coefficient 2 (a2)	-1.25 +/- 1.44
Scatter in the residuals of the lightcurve fit is	49.74 %
Best Comparison Star	#1 - [65, 369]
Optimal Aperture	3.01
Optimal Annulus	11.59
Transit Duration (day)	0.032 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.058 vs 0.1594 (deviation 0.46σ)
Duration fit vs published	0.032 d vs 0.0512 d (deviation 1.21σ)
Mid-time O–C	-5.8 min
Depth SNR	2.4
Residual scatter	49.74 %

Light curve

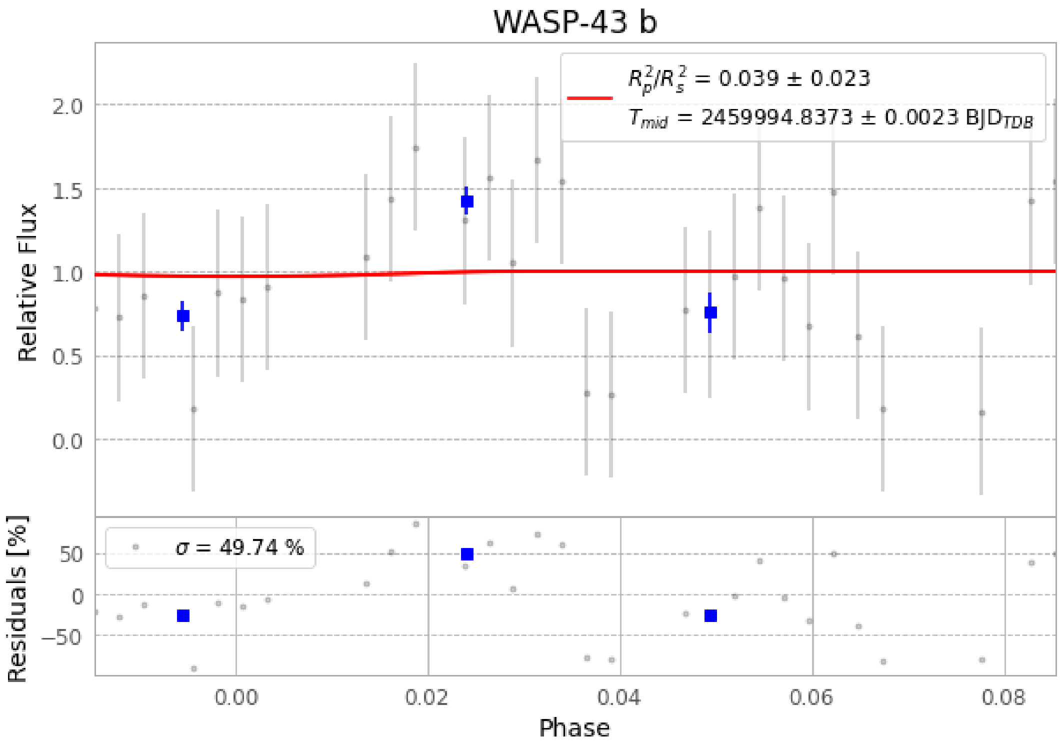

Figure 1 — FinalLightCurve_WASP-43 b_2023-02-19.png (SHA-256 b627567546de49b4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 198], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8e6bf496efa24375…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

41 FITS frames (27 MB), `WASP-43230219073915.FITS` → `WASP-43230219093916.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230219.log (6a9720750a03…), FinalLightCurve_WASP-43 b_2023-02-19.png (b627567546de…), FinalLightCurve_WASP-43 b_2023-02-19.csv (f005bce900c6…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-25 03:09Z.